

Shivalik HEPCATS Summer 2026

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6. Do you want to present your work? *

☒ Yes☐ No

7. Title of talk *

Zoom-whirl orbital dynamics and kludge gravitational waveforms in extreme mass-ratio inspirals near charged black holes surrounded by perfect fluid dark matter

8. Abstract for the talk *

With the recent advancement in gravitational wave studies, more focus has been shifted to theoretically studying the extreme mass ratio system due to its application in detecting gravitational waves by future space-based observatories. The periodic zoom-whirl trajectory of a celestial body with mass m near a black hole with mass M , where $m \ll M$, has its correspondence with a specific structure of gravitational waveform. This paper scrutinises the analysis of zoom-whirl orbits around a supermassive black hole containing charge Q , manifested in the surroundings of perfect fluid dark matter (PFDM), and their association with gravitational waves. We analysed the timelike geodesic and explained the MBOs and ISCOs. The finding describes that the implementation of the PFDM parameter provides a unique character to the charged black hole. The zoom-whirl periodic orbits bounded by the energy and angular momentum of respective MBOs and ISCOs are revealed to be significantly influenced by the existence of the charge (Q) and the PFDM (γ) parameters. Further, we computed the gravitational waves, using the Kludge's waveform model, for periodic orbits exhibiting the zoom and whirl essence in waveforms. Our analysis explains the significance of Q and γ parameters in periodic orbits as well as corresponding gravitational waveforms.



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